First Task Multivariate Analysis

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1 Introduction

The original dataset **`all_seasons.csv`** used for this assignment was found on *Kaggle*, it contains over two decades of data on each player who has been part of an NBA teams' roster, leading to nearly 13.000 rows. The raw dataset consisted of 22 features, which included physical attributes (age, height, weight), career information (team, draft year, draft round) and some basic game statistics (average points, rebounds, assists) among others.

In order to reduce the number of observations, our analysis will focus exclusively on the 2022-23 season. Similarly, to reduce the number of features, we decided to keep the following ones:

- `player_name` : Name of the player
- `team_abbreviation` : Abbreviated name of the team the player played for (30 different teams)
- `player_age`: Age of the player
- `player_height`: Height of the player in cm
- `player_weight`: Weight of the player in kg
- `country`: Name of the country the player was born in

The NBA Draft is an annual event that takes place before the season starts, allowing teams to select new players to join the league. This process is divided into 2 rounds, from all candidates just 60 players are finally chosen. It is the way how players with no prior experience coming from college or international leagues enter the NBA. With this in mind, we use the variables:

- `draft_year`: The year the player was drafted
- `draft_number`: The number at which the player was picked in his draft round
- `gp`: Games played throughout the season
- `pts`: Average number of points scored
- `reb`: Average number of rebounds grabbed
- `ast`: Average number of assists distributed
- `ts_pct`: Measure of the player's shooting efficiency that takes into account free throws, 2 and 3 point shots.

2 Data preprocessing

In order to meet the specifications for the mixed-type multivariate dataset we needed to do some data preprocessing. For example, after filtering the data by the most recent season, we obtained a total of 539 players. However, in order to make use of the `draft_year` variable, it was necessary to handle the presence of *Undrafted* players. As a result, the 134 players labeled as *Undrafted* were removed. The next step was to analyse the `draft_number` feature. We identified 3 additional players who still had an *Undrafted value*, meaning that although they had an assigned `draft_year`, they were ultimately not selected in the draft. Consequently, these players were also removed, resulting in a clean dataset of 402 players with no NA values.

2.1 Binary variables

- `from_US`: this feature was created from the `country` variable, where 1 indicates if the player was born in the US and 0 otherwise
- `is_tall`: this feature was derived from the `player_height` variable, where a value of 1 indicates that the player's height is greater than the median, and 0 otherwise

2.2 Multiclass categorical variables

- `draft_number`: was originally a discrete quantitative variable, but it was transformed into a multiclass categorical feature called `draft_number_cat`, by splitting it into quartiles, resulting in four groups: Veteran (0–25%), Prime (25–50%), Early Prime (50–75%), and Young (75–100%). We interpret this variable as an indicator of tenure, meaning that players from earlier draft years have more experience in the league.
- `pts`: was originally a continuous quantitative variable, but it was transformed into a qualitative ordinal variable (`sc_level`) using tertiles, resulting in three levels: Low (0–33%), Medium (33–66%), and High (66–100%).

3 Data description

3.1 Summary

We begin our exploratory data analysis with a summary statistics of the **quantitative variables**, Table 1 shows that they have very different scales. This was expected, as our features are measured in very different units (kg, cm, points per game, percentages, etc.). For example, points, rebounds and assists vary on a much larger scale than variables such as true shooting percentage or age.

Table 1: Summary of quantitative variables

Statistic	age	player_height	player_weight	draft_number	gp	pts	reb	ast	ts_pct
Min.	19.00	180.3	74.39	1.00	1.00	0.000	0.000	0.000	0.0000
1st Qu.	23.00	193.0	90.72	9.00	39.00	5.025	2.200	0.900	0.5310
Median	25.00	200.7	97.52	20.00	59.00	8.800	3.500	1.500	0.5725
Mean	26.05	199.9	98.40	22.59	53.11	10.488	3.969	2.354	0.5680
3rd Qu.	29.00	205.7	106.48	34.00	71.00	14.300	5.000	3.375	0.6100
Max.	39.00	221.0	131.54	60.00	83.00	33.100	12.500	10.700	0.9000

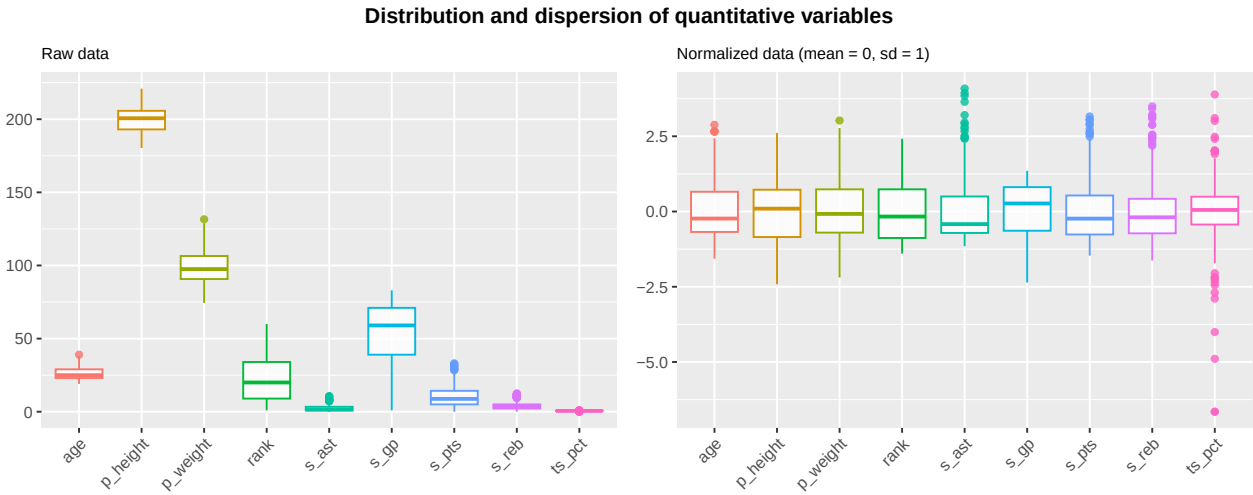
Meanwhile, the summary statistics for the **qualitative variables**, shown in Table 2, reveal a a highly balanced league in terms of `draft_year_cat` (ie years of experience) and `sc_level`. Regarding nationality, about 75% of players come from the US, while nearly 60% of the players are consider to be shorter in terms of the `is_tall` variable.

Table 2: Summary of categorical variables

draft_year_cat	sc_level	from_US	is_tall
Young : 98	Low :134	0:101	0:237
Early Prime: 87	Medium:132	1:301	1:165
Prime :110	High :136		
Veteran :107			

3.2 Boxplots

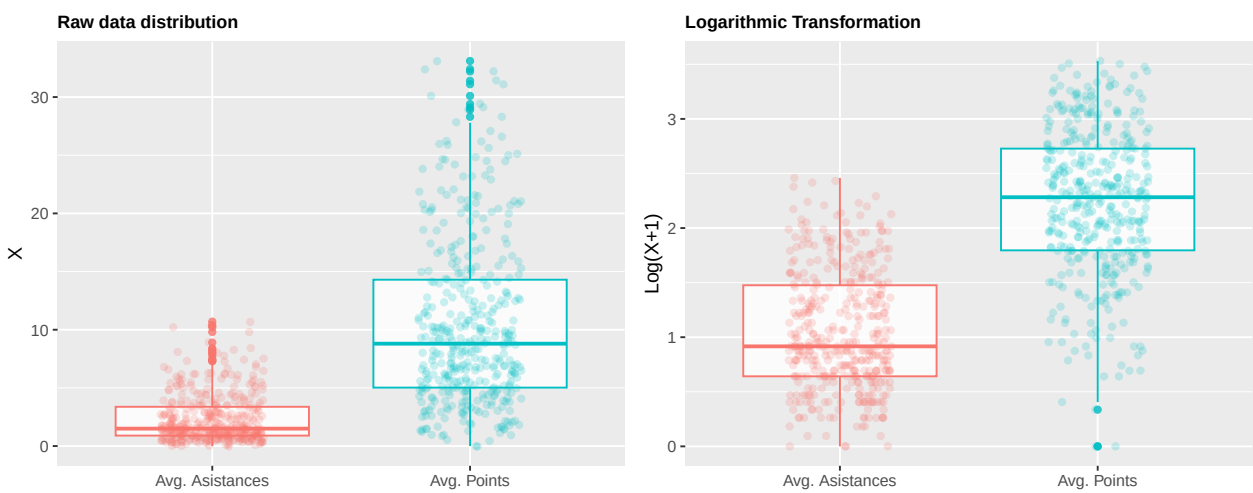
Recall the problem about variance comparison across quantitative variables, as can be seen in the left-hand side plot. To adress this issue normalization is applied, resulting in the right-hand plot. Notice that this process does not change the shape of the distributions, only the scale of the data. Therefore, patterns in the original data, such as skewness, dispersion, or outliers are preserved, while enabling meaningful comparison between different variables.



In particular, we can observe a small group of players, which could be potentially considered as elite players, exceeding the upper whiskers of the boxplot in variables such as points, assists and rebounds.

These outlier points likely represent the superstars, whose performance is far above the average. Due to the considerable difference between an average player and an elite one, the distribution is naturally right-skewed. We could correct this issue by applying the $\log(x + 1)$ transformation, we need to add 1 to avoid issues with players who recorded neither assists nor points during the season.

The resulting distribution becomes much more closer to normal. This transformation mitigates the influence of extreme values and significantly reduces the initial skewness, as can be seen in the following pictures, although it is not strictly necessary for the later analysis.



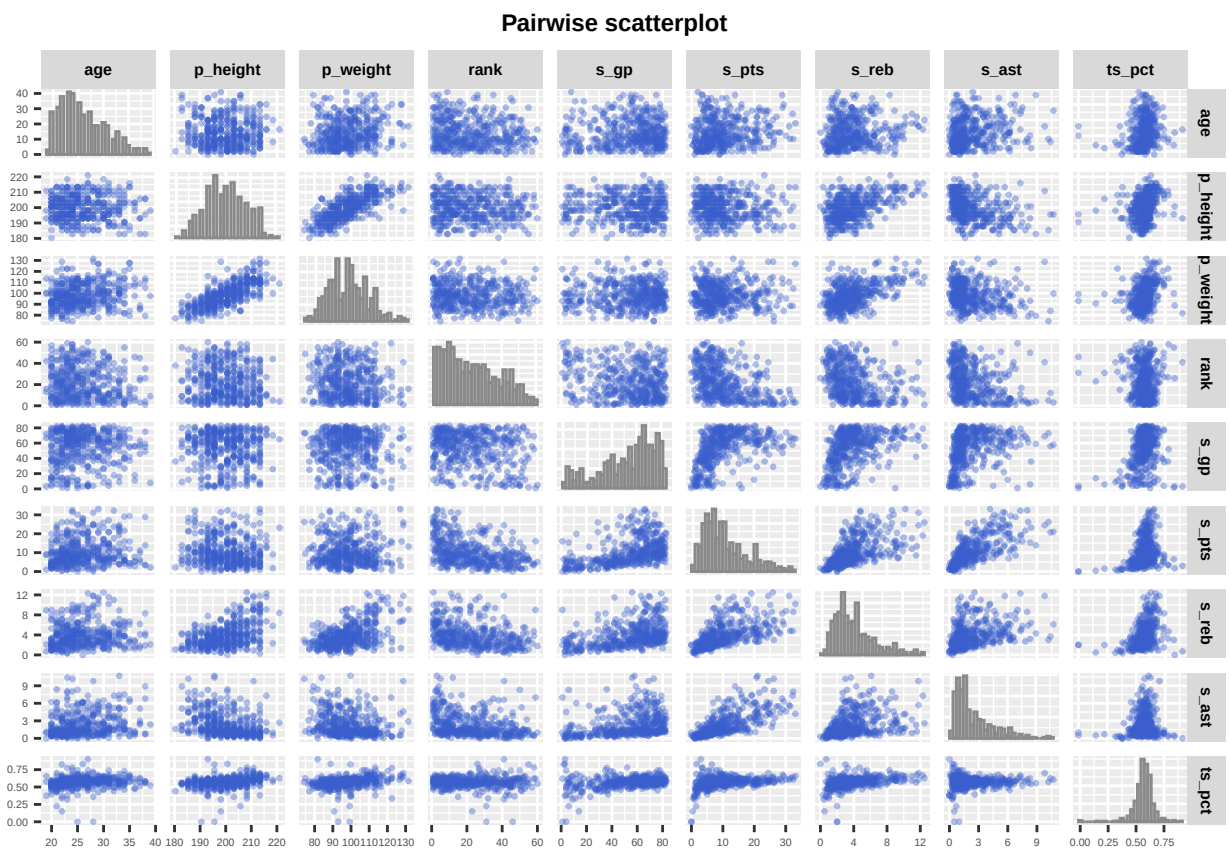
To look for possible correlation between quantitative variables in our dataset, we generate a pairwise scatterplot. At first glance, a strong **positive linear correlation** highlights between players' anthropometric characteristics, specifically **height and weight**. There is also a moderate **positive correlation between height and rebounds**. This is expected in basketball, as the pivot is usually the tallest and heaviest player, which is responsible for collecting rebounds. Conversely, the opposite pattern is observed with assists: taller players tend to have fewer assists.

3.3 Scatterplots

We will now refer to the group of variables related to a player's number of points, rebounds, and assists as their *performance metrics*. By this approach, we can obtain the following insights:

- *Games played and Performance metrics*: Across all combinations, an increasing trend is observed, indicating that players with more games played generally obtain higher values in their statistics. However, variability increases for players who participate in more than 40 games during the season.
- *Rank and Performance metrics*: The variable rank shows a weak negative and kind of non linear correlation with all performance statistics. While it does not fully determine a player's productivity, a clear pattern highlights: players selected in the last positions of the draft (ie high values for draft rank) tend, on average, to perform worse during the season.
- *Performance metrics*: In the scatterplots relating performance metrics, what stands out the most is the funnel-shaped pattern and its width, which means that there is some *heterocedasticity* between these variables. There is also a positive association between them. In other words, players with higher values in one metric generally tend to score higher in others, although this relationship might not be linear.

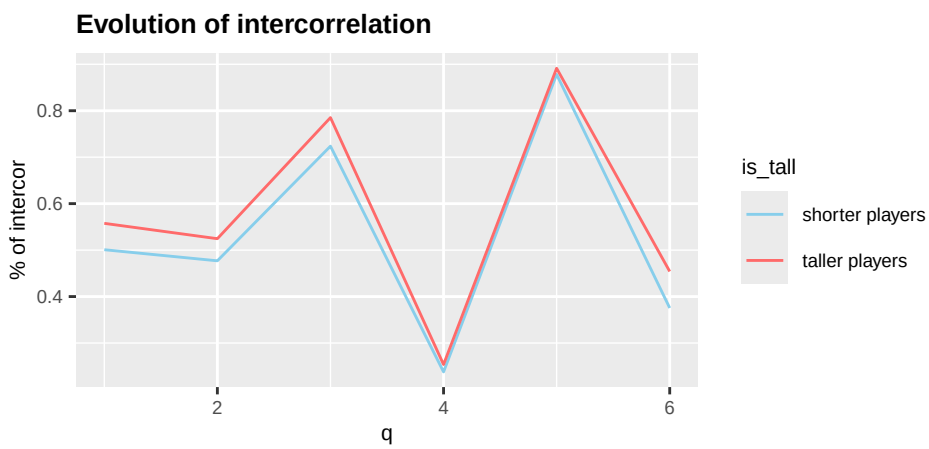
Shooting efficiency is remarkably consistent throughout the league, with most players ranging between 45% and 65%. When compared with other variables, no clearly pattern or correlation is observed. Similarly, the remaining variables show no clear structure or relationship in the data.



We can also compute the six indexes q_i to measure intercorrelation among our variables. Recall that Pearson correlation coefficient only detects linear relationships, and the same applies to q:

$$q = [0.469, 0.547, 0.841, 0.187, 0.896, 0.466]$$

Low values in q_1, q_4, and q_5 may be due to the non-linear relationships previously observed in the scatterplot. We also decided to check if the intercorrelation among all quantitative variables (excluding numeric height) was stable across the `is_tall` binary variable. Overall, the correlations tend to be consistently higher in the taller-player (coded as 1) group than in the shorter-player group (coded as 0).



$$q_{tall} = [0.53,\ 0.52,\ 0.82,\ 0.22,\ 0.90,\ 0.45] \quad q_{short} = [0.50,\ 0.48,\ 0.72,\ 0.24,\ 0.88,\ 0.38]$$

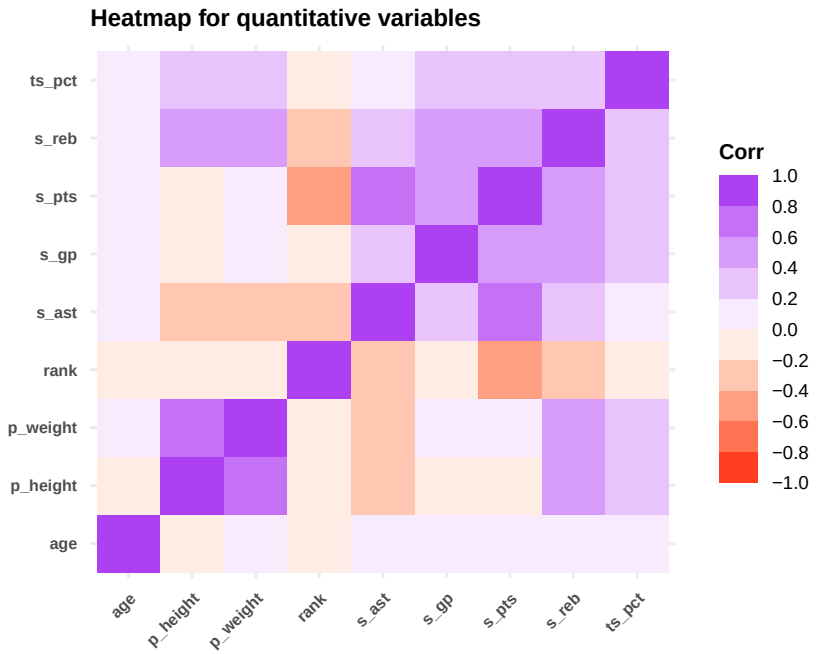
3.4 Heatmap

In the heatmap, the correlation coefficient between `height` and `weight` appears to be the strongest positive one (approximately 0.73), which is consistent with the pattern previously observed in the scatterplot. A similar situation occurs with the correlation between the number of `rebounds` and the player's anthropometric characteristics, which is moderate (around 0.5). Finally, there is a weak negative correlation (approximately –0.3) between the number of `assists` and variables such as `height`, `weight`, and `rank`. In addition, the correlation between `rank` and `points` is slightly stronger, at around –0.4

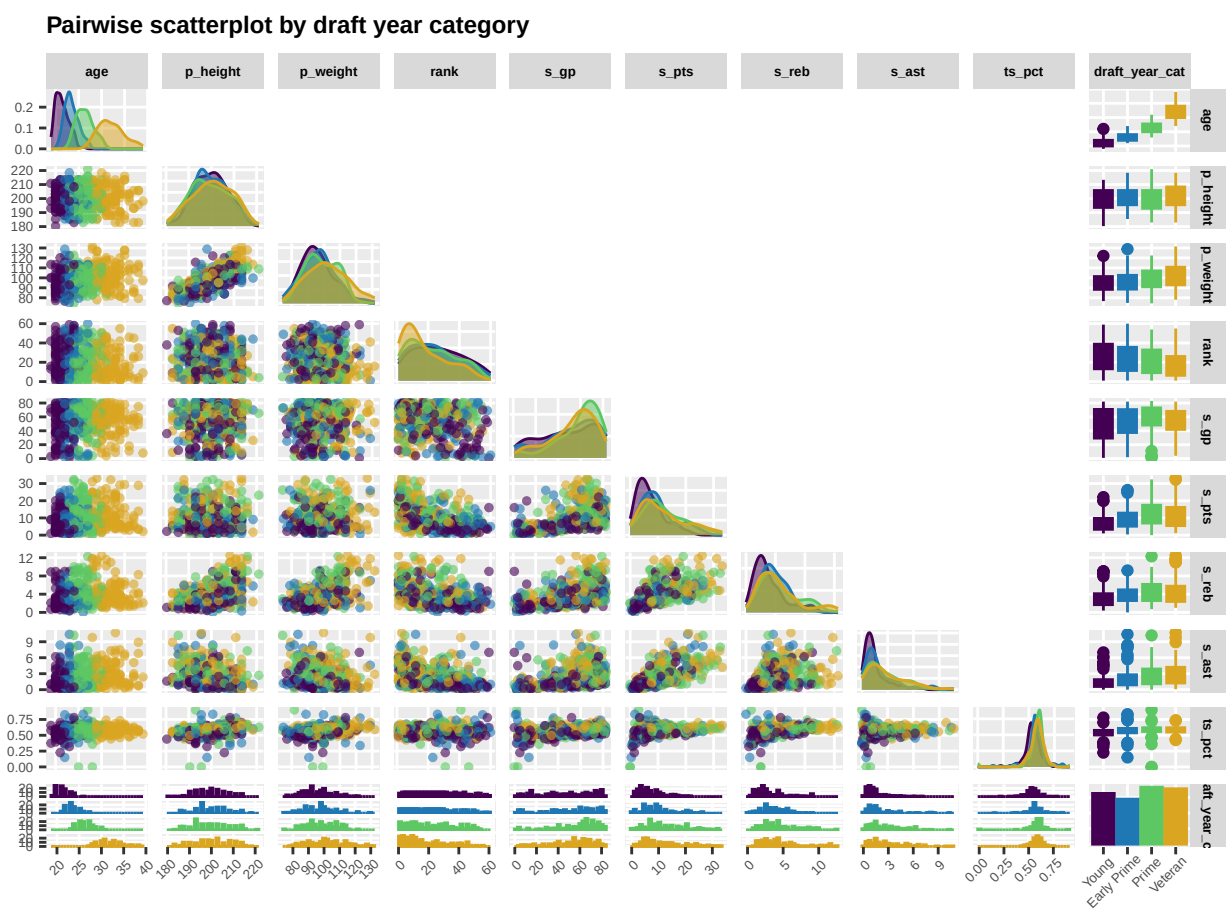
3.5 Conditional scatterplots

When conditioning the previuos pairwise scatterplot on the categorical variable `draft_year`, an obvious pattern emerges in the first column `Age`. As expected, players classified as Veteran, those with more years of experience in the league, are also the oldest ones.

If we focus on the performance metrics (points, rebounds, assists) and shoot efficiency, we appreciate that the least experienced players, those who have recently joined the NBA (represented by purple and blue points) are predominantly concentrated in the lower ranges of these distributions. In contrast, players in the *Prime and Veteran* categories (green and yellow) tend to take higher values for these metrics. This pattern is consistent with the boxplots where both, quartiles and averages increase progressively with player seniority.



Moreover, rookies (non-experience players) show a substantial variability in the number of games played, reflected in the length of the purple boxplot. This stands in contrast to the *Veteran* group, whose boxplot reflects lower dispersion, indicating a more regular and constant level of participation throughout the season.



We may conclude that players either tend to improve their performance as they gain experience, or those who fail to improve their skills are released from the league before reaching the *Veteran* category. Meanwhile, the *Young* category includes both players who will eventually become top performers and those who are likely to struggle, and at this early stage it is difficult to distinguish between them.

4 Principal Component Analysis

In this part of the work, we focus on the quantitative variables of the dataset to explore their multivariate structure using PCA. The idea is to reduce the number of variables while keeping as much information as possible, and to see whether we can identify meaningful patterns in player performance during the 2022–23 NBA season.

For this analysis we only use the numerical variables (age, height, weight, draft number, games played, points, rebounds, assists and true shooting percentage). Categorical variables such as position or nationality are not included here.

Recall the problem seen in the summary statistics and the boxplots for the quantitative variables, they have very different scales and dispersions. Because PCA is based on variances, variables with larger variance would dominate the first principal components if we worked with the covariance matrix. To avoid this issue and give all variables a comparable weight, in the following we perform PCA on the correlation matrix R instead of on the covariance matrix R.

4.1 Extraction of the principal components

We can now perform Principal Component Analysis in our data.

```
R <- cor(X)

# Eigendecomposition of R
R_eig  <- eigen(R)
lambda <- R_eig$values    # eigenvalues
T_mat  <- R_eig$vectors   # eigenvectors (loadings)

# Percentage of variance explained

var_explained <- lambda / sum(lambda) * 100
```

We obtain:

Table 3: Eigenvalues and percentage of variance explained (manual computation)

Component	Eigenvalue	% of variance
PC1	2.944	32.71

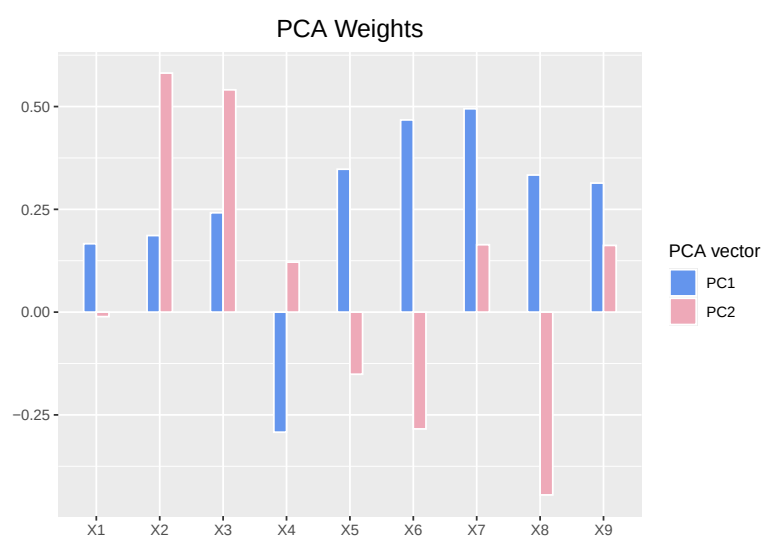
PC2	2.161	24.01
PC3	1.018	11.31
PC4	0.981	10.9
PC5	0.662	7.36
PC6	0.578	6.42
PC7	0.248	2.76
PC8	0.213	2.36
PC9	0.195	2.17

Notice that we can get the same result using the built-in function `prcomp`.

Table 4: Eigenvalues and explained variance from prcomp

	eigenvalue	variance.percent	cumulative.variance.percent
Dim.1	2.9440588	32.711765	32.71176
Dim.2	2.1612536	24.013929	56.72569
Dim.3	1.0177692	11.308547	68.03424
Dim.4	0.9810537	10.900597	78.93484
Dim.5	0.6621026	7.356695	86.29153
Dim.6	0.5775395	6.417105	92.70864
Dim.7	0.2482629	2.758476	95.46711
Dim.8	0.2125814	2.362015	97.82913
Dim.9	0.1953784	2.170871	100.00000

At this point, we can analyse the eigenvectors in order to give a meaning to the new dimensions $Y_1, Y_2, \ldots$.

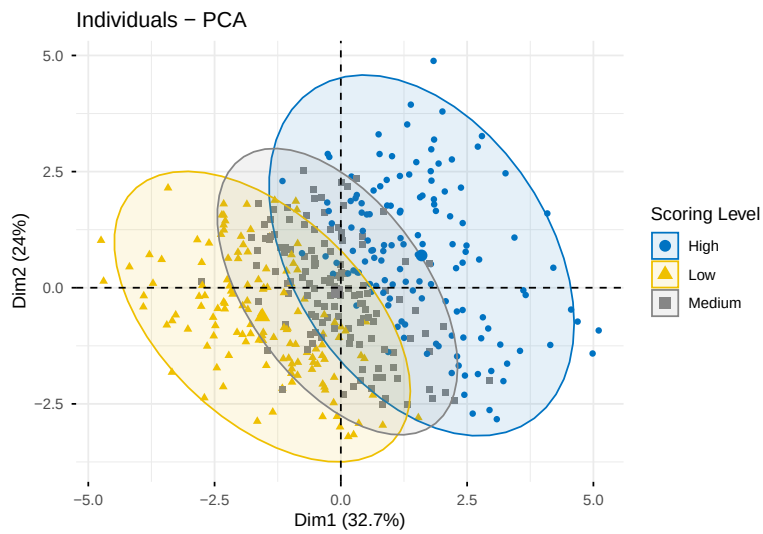


In this graphic, we can observe that PC1 weights of gp (number of games played) and pts (average points of the player) are positive and relatively large and draft number weight stands out as the

only negative weight. This means that players with a high score on PC1 tend to play more games and score more points, also corresponding to better draft positions (low draft number). A possible interpretation of this is PC1 as a on.court contribution. A high PC1 implies a high-impact player

On the other hand, PC2 has strong positive weights on player_height and player_weight, while it has a negative large weights on ast (assited) and a smaller contribution of pts. So, players with high PC2 tend to be taller and heavier, but usually score and assist less and are less involved in playmaking. Players with low PC2 are the opposite: smaller, lighter players who score and assist more. Taking this into account, we can interpret PC2 as a role component that separates interior players (bigs) on the positive side from perimeter players or guards on the negative side.

We now project the players onto the first two principal components to see whether the reduced space reveals meaningful groups, especially in terms of scoring level and player role.

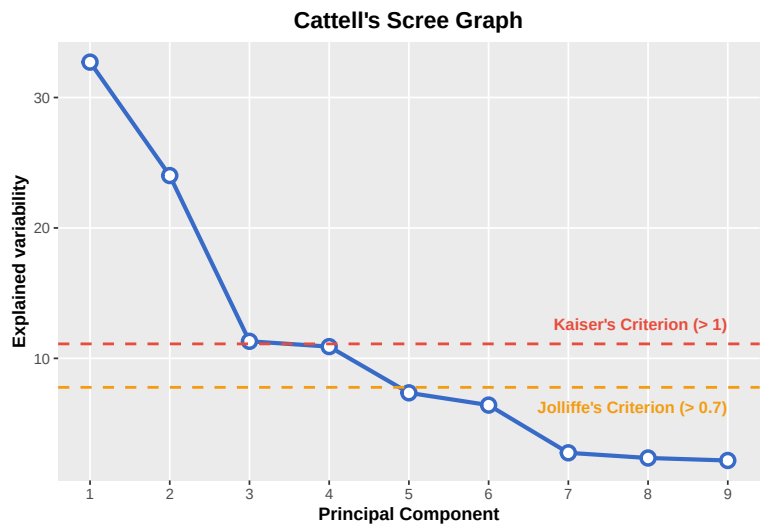


The projection of the players onto the first two principal components shows a clear structure. Along the first component (Dim1), players are mostly ordered according to their scoring level: "High" scorers appear on the right-hand side of the plot, while "Low" scorers are located on the left. This is consistent with the interpretation of PC1 as a general performance component, since players who score more points also tend to play more minutes and contribute more in rebounds and assists. PC2 mainly distinguishes taller, interior-oriented players from smaller perimeter players.

It is worth to mention that the first two principal components explain 56.7% of the total variance, which is enough to reveal clear trends but not to fully separate the groups. This, together with the inherent noise in player performance, explains the overlap between scoring levels in the PCA plot.

4.2 Methods for Discarding Components

In this subsection we address the classical question of how many principal components should be retained. We apply the five criteria discussed in class: cumulative percentage of explained variance, Kaiser's rule, Jolliffe's modification, Cattell's scree plot and the dimensionality Test .



The eigenvalues of the correlation matrix show that the first component explains about 32.7% of the total variance, the second around 24%, and the third and fourth roughly 11% each. Using the cumulative percentage of explained variance, we need the first four components to exceed the 70% threshold, so this criterion suggests retaining $\mathbf{k = 4}$ components.

Kaiser's rule (keep components with eigenvalues larger than 1) recommends retaining the first 3 components ($k = 3$). Jolliffe's more permissive criterion (eigenvalues larger than 0.7) leads again to four components ($k = 4$). The scree plot displays a clear drop between PC1 and PC2 and a noticeable change of slope after PC3, indicating that most of the information is concentrated in the first few components and that keeping between three and four components is reasonable.

The last criterion is the dimensionality test:

Table 5: Dimensionality Test Results (H0: Last p-k eigenvalues are equal)

k	Statistic	df	p_value	Decision
7	0.707	2	0.70228	Fail to Reject H0
6	5.923	5	0.31381	Fail to Reject H0
5	166.249	9	0.00000	Reject H0 (Keep more)
4	272.859	14	0.00000	Reject H0 (Keep more)
3	454.644	20	0.00000	Reject H0 (Keep more)
2	569.246	27	0.00000	Reject H0 (Keep more)
1	1008.581	35	0.00000	Reject H0 (Keep more)
0	1464.300	44	0.00000	Reject H0 (Keep more)

The test for the equality of the last $p - k$ eigenvalues fails to reject H_0 for $k = 6$ and $k = 7$, which would formally suggest keeping at least six components. However, it is known that this test tends to be very sensitive with large samples such as ours (more than 400 players) and tends to overestimate the dimensionality with real data.

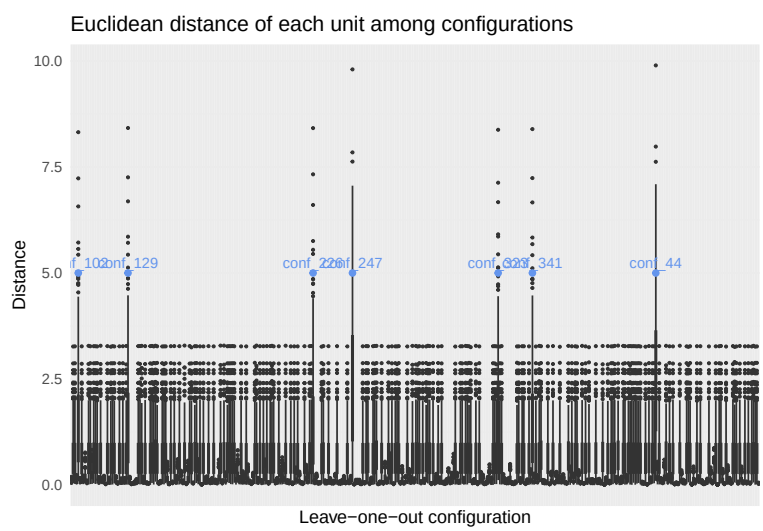
Taking all criteria together, and also considering interpretability, we decide to retain four principal components. Once the number of retained components has been determined, the PCA solution can be used to approximate the original data and to perform further analyses. The first four components already preserve more than 70% of the total variance, meaning that most of the relevant information is contained in the reduced representation.

4.3 Stability Analysis

In this final step, we assess the stability of our PCA solution. It is important to verify that the principal components are not driven by a single influential observation. To do so, we perform a jackknife leave-one-out procedure: we recompute the PCA n times, each time removing one player from the dataset, and we measure how much the configuration of individuals changes.

Table 6: Global summary of stability radius

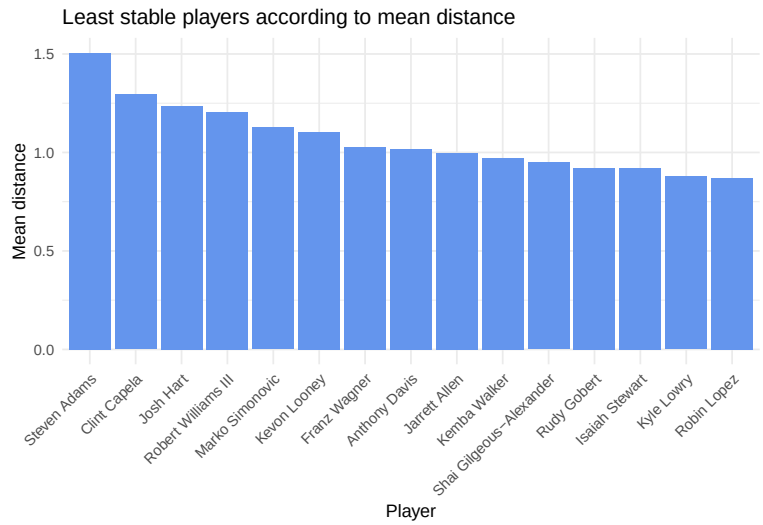
Statistic	min	max	q50	q75	q90	q95
mean	0.005	2.990	0.133	0.697	0.716	0.735
sd	0.002	1.584	0.066	0.545	0.538	0.530
median	0.005	2.753	0.122	0.550	0.563	0.583
mad	0.001	1.064	0.039	0.313	0.303	0.293

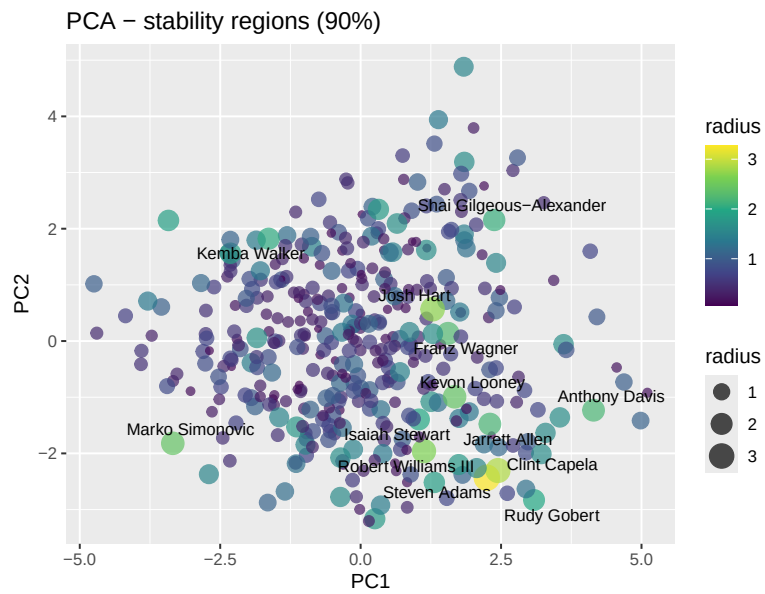


The results show that the overall PCA configuration is highly stable. For most players, the Euclidean distance between the original scores and the jackknife configurations remains very small, meaning that removing any single observation does not substantially modify the two-dimensional representation.

The boxplots of the leave-one-out distances confirm that the vast majority of configurations remain tightly clustered, while a small number of removals produce larger deviations. The stability maps reinforce this result: most players appear with very small circles, whereas a few units have large 90% regions and are therefore more influential in defining the PCA axes. These units correspond to

extreme profiles in the dataset, typically players with very high performance values or unusual statistical combinations.





This plot projects the players onto the first two principal components and visualizes the stability of their positions. The color and size of each point represent the 90% stability radius: larger and brighter points correspond to players whose coordinates vary more when other individuals are removed from the analysis. Most players show very small stability regions, meaning that their position in the PCA space is highly robust. Only a small group displays large radius, indicating that they are influential observations.

The two graphics are consistent. We can see that players with the highest mean distances in the jackknife analysis are the same ones that display the largest stability radius in the PCA map.

Interestingly, several of the least stable players identified (e.g., Steven Adams, Clint Capela, Rudy Gobert or Anthony Davis) are well-known impactful players in the NBA, often dominating in

rebounding or overall performance. Their extreme statistical profiles place them at the edges of the PCA space, which explains why the components are more sensitive to their presence.

Overall, the stability analysis indicates that the PCA solution is robust: the main components and the relative positioning of the players do not depend on any single observation. This strengthens the validity of the interpretations presented in the previous sections.

5 Conclusions

In this work we applied Principal Component Analysis to a dataset of NBA players from the 2022–23 season with the aim of reducing dimensionality and identifying the main patterns behind player performance. The first two components captured the most relevant structure in the data: PC1 reflected a general performance axis (scoring, rebounds, assists and playing time), while PC2 distinguished between bigger interior players and perimeter-oriented guards. This two-dimensional representation already revealed meaningful groupings, particularly in terms of scoring level and player role.

Using several criteria for component selection, we concluded that retaining four principal components provides a good balance between explained variability and interpretability, preserving more than 70% of the total information. The remaining components mostly contained noise or very specific variation.

Finally, the stability analysis showed that the PCA solution is robust: the positions of most players remained stable under jackknife resampling, and the interpretation of the components did not depend on any single observation. Only a small number of influential players—often high-impact or statistically extreme profiles—had a noticeable effect on the configuration, which is consistent with their dominant roles in the league.